

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

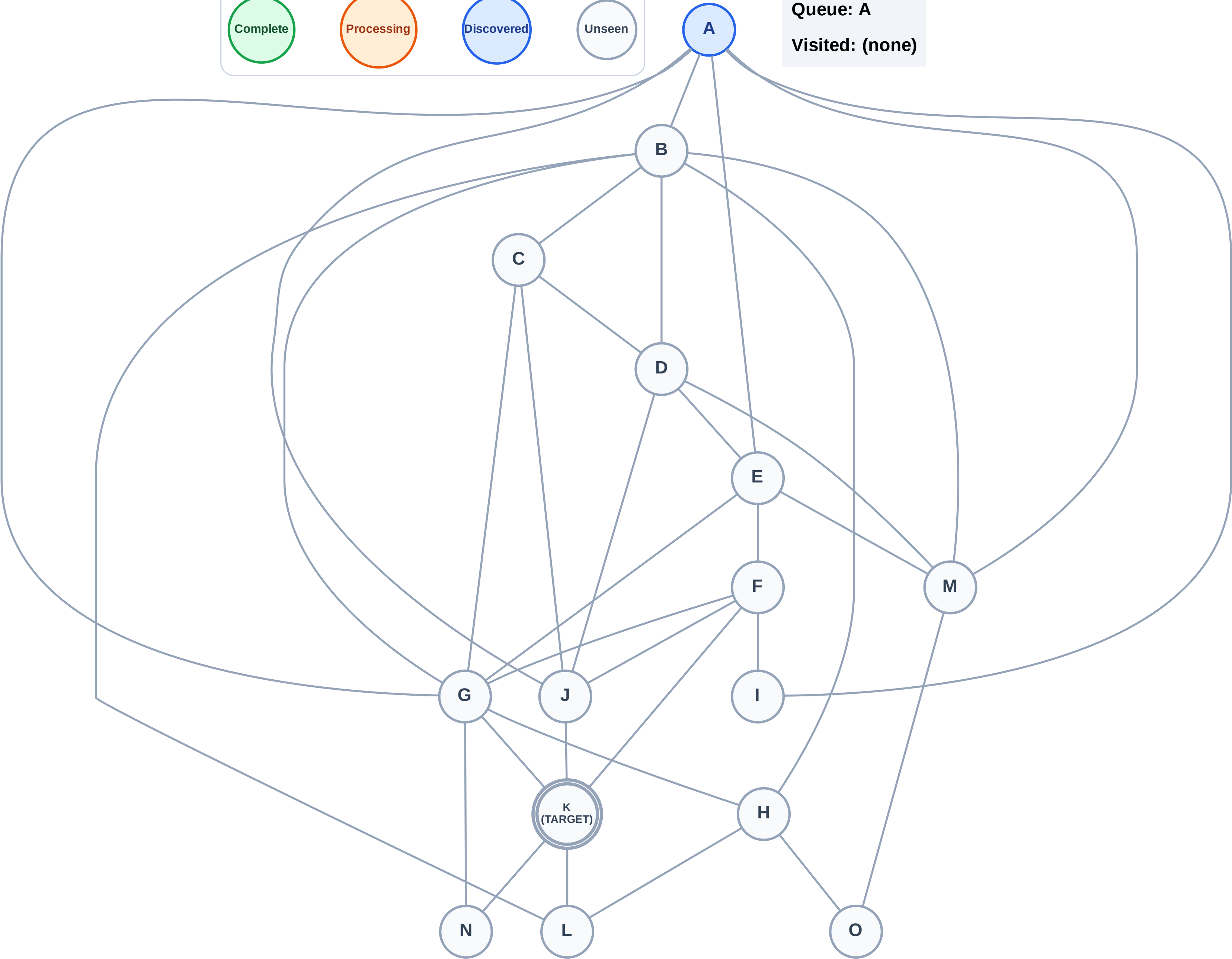
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

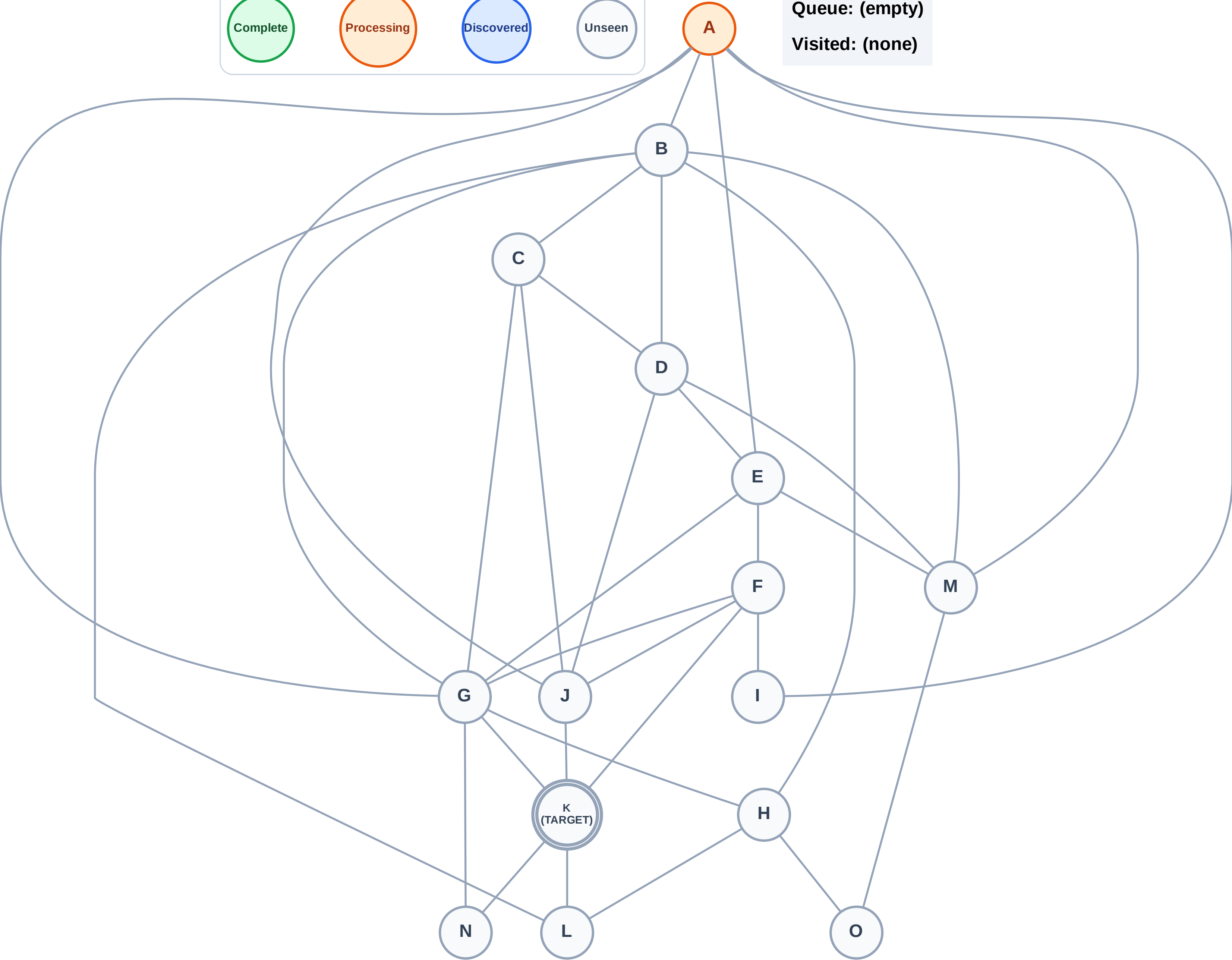
Complete

Processing

Discovered

Unseen

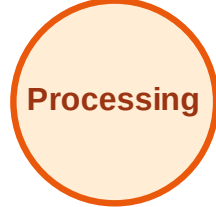
Queue: (empty)
Visited: (none)



BFS Traversal

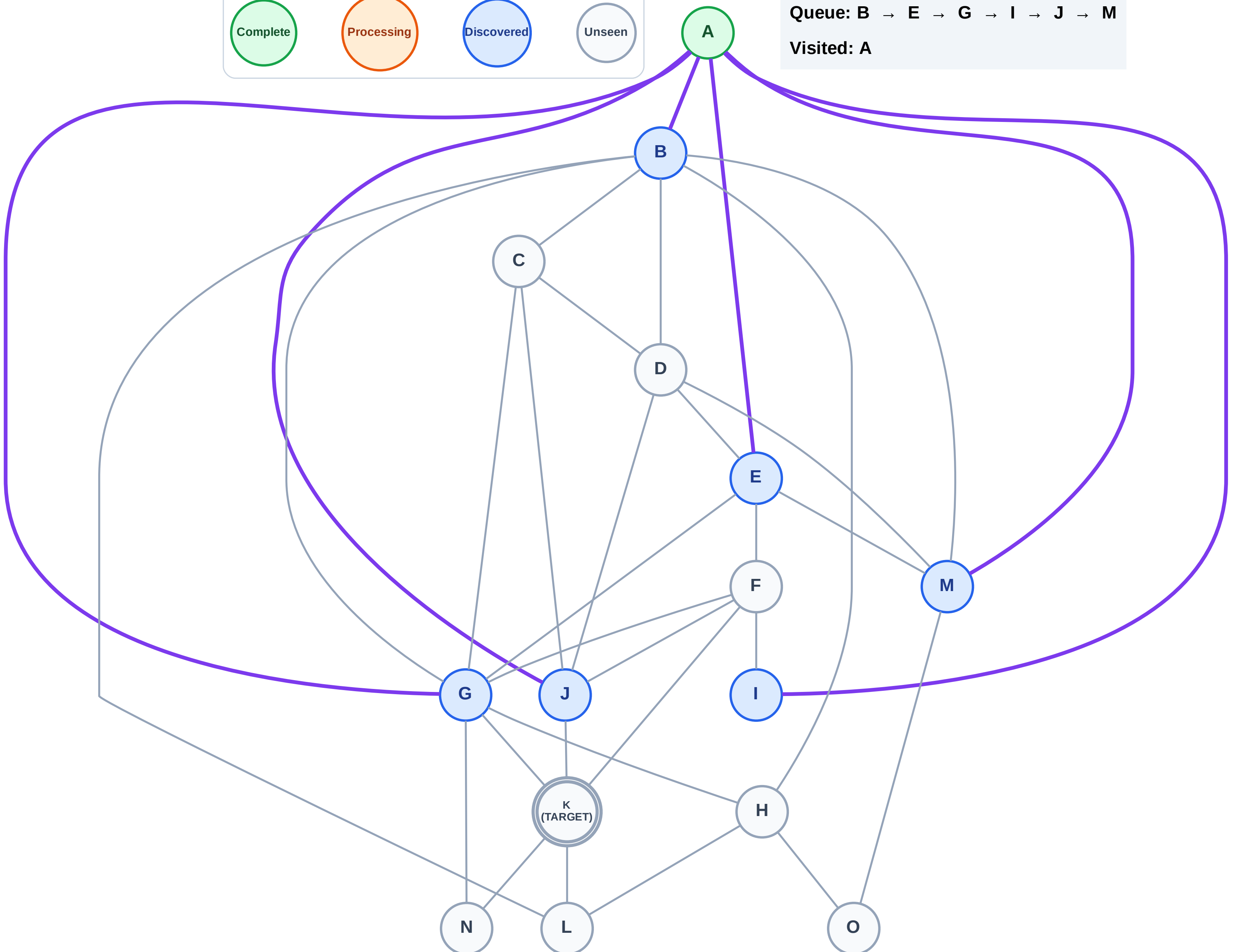
Step 3: Discover B, E, G, I, J, M from A

Legend



Queue: B → E → G → I → J → M

Visited: A



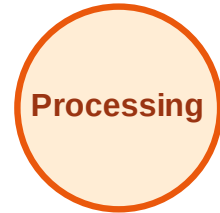
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



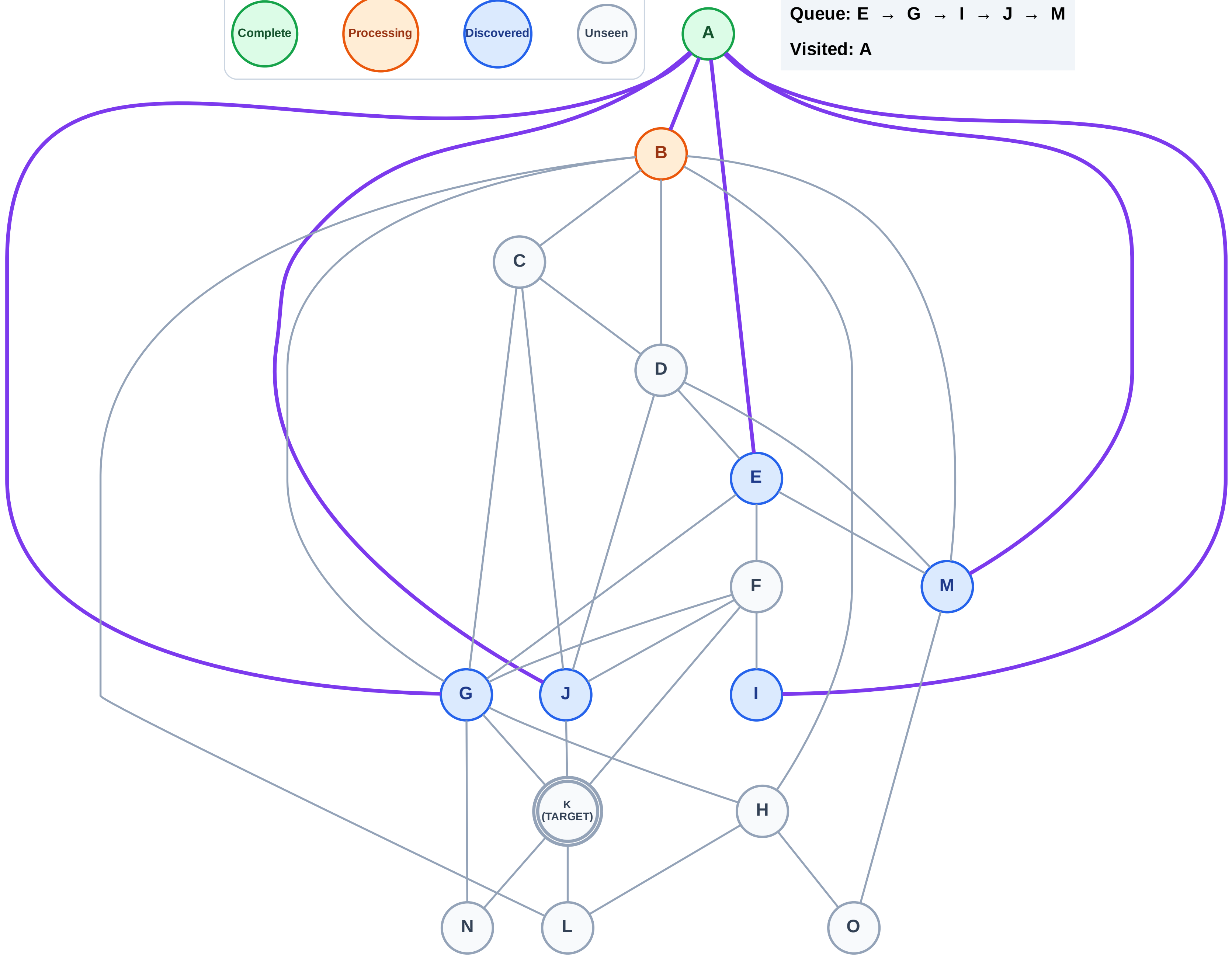
Discovered



Unseen

Queue: E → G → I → J → M

Visited: A



BFS Traversal

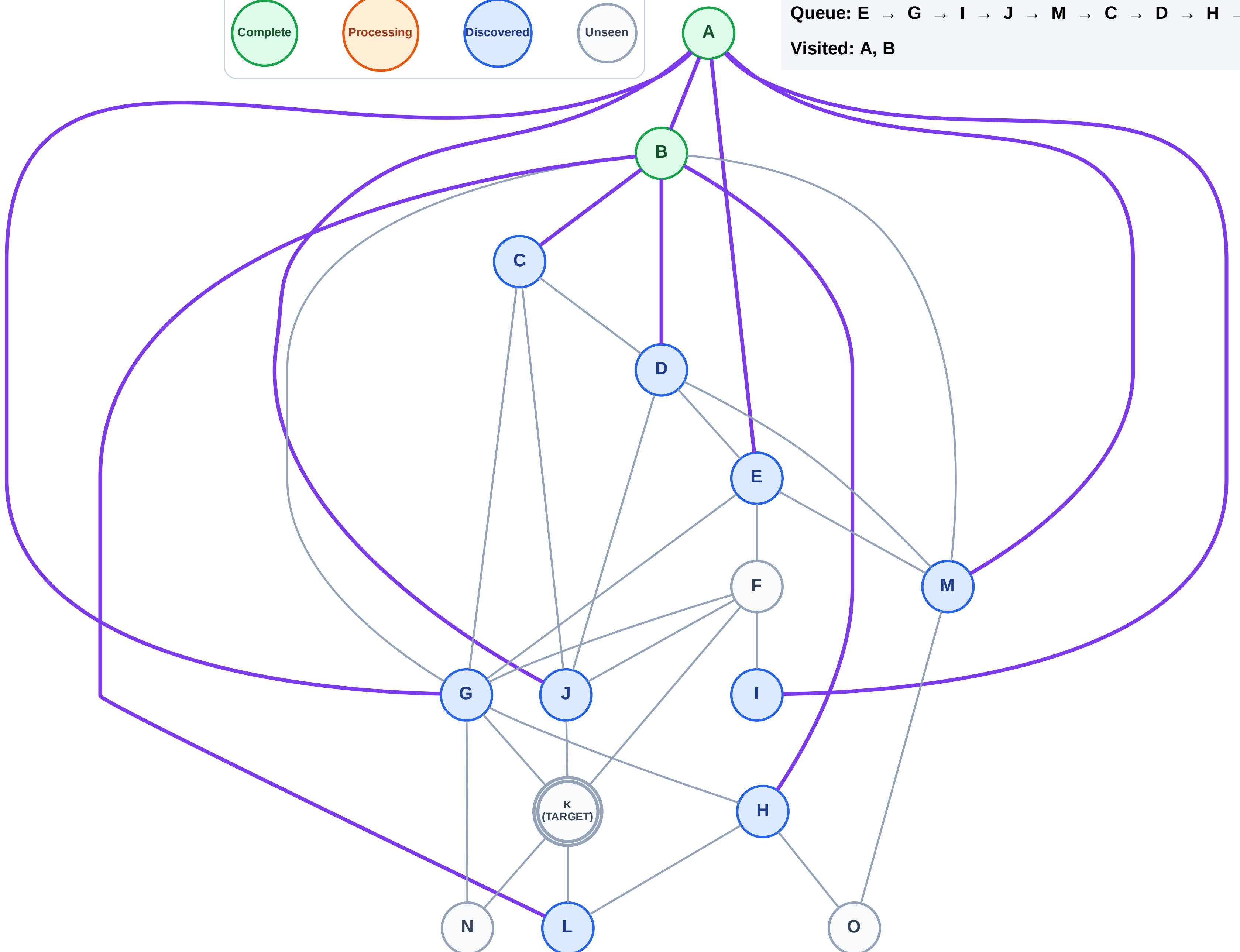
Step 5: Discover C, D, H, L from B

Legend



Queue: E → G → I → J → M → C → D → H → L

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

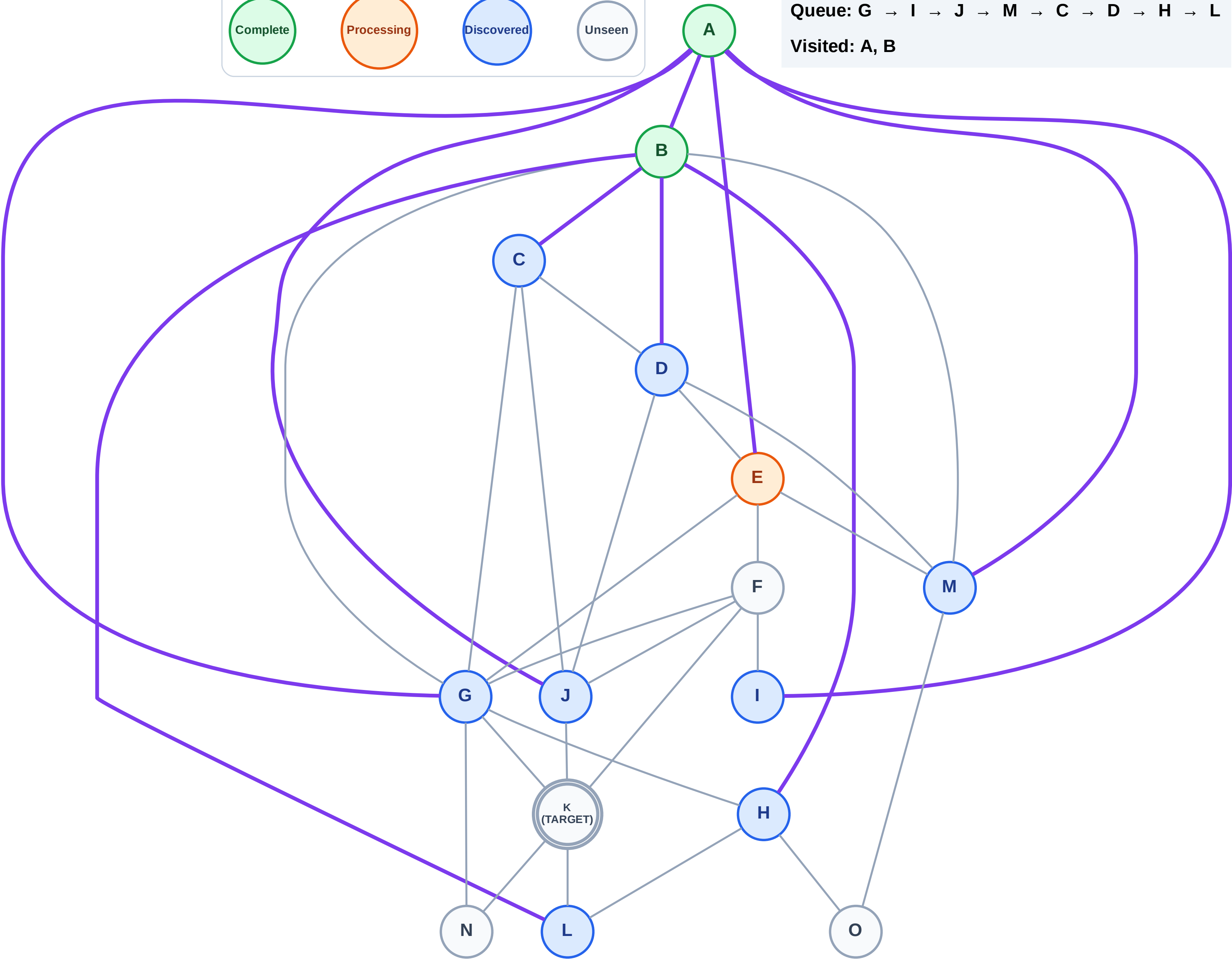
Processing

Discovered

Unseen

Queue: G → I → J → M → C → D → H → L

Visited: A, B



BFS Traversal

Step 7: Discover F from E

Legend



Complete



Processing



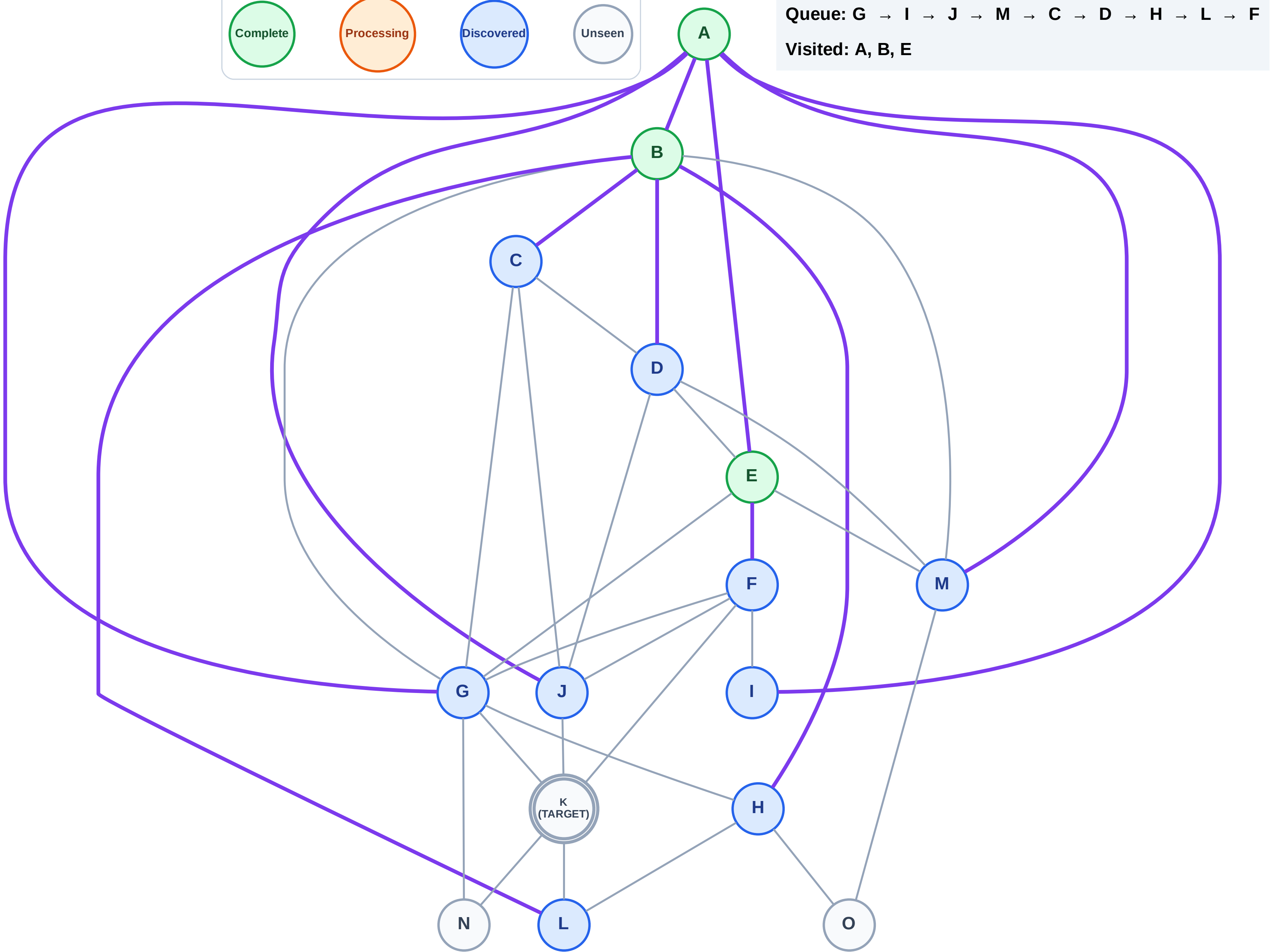
Discovered



Unseen

Queue: G → I → J → M → C → D → H → L → F

Visited: A, B, E



BFS Traversal

Step 8: Process node G

Legend

Complete

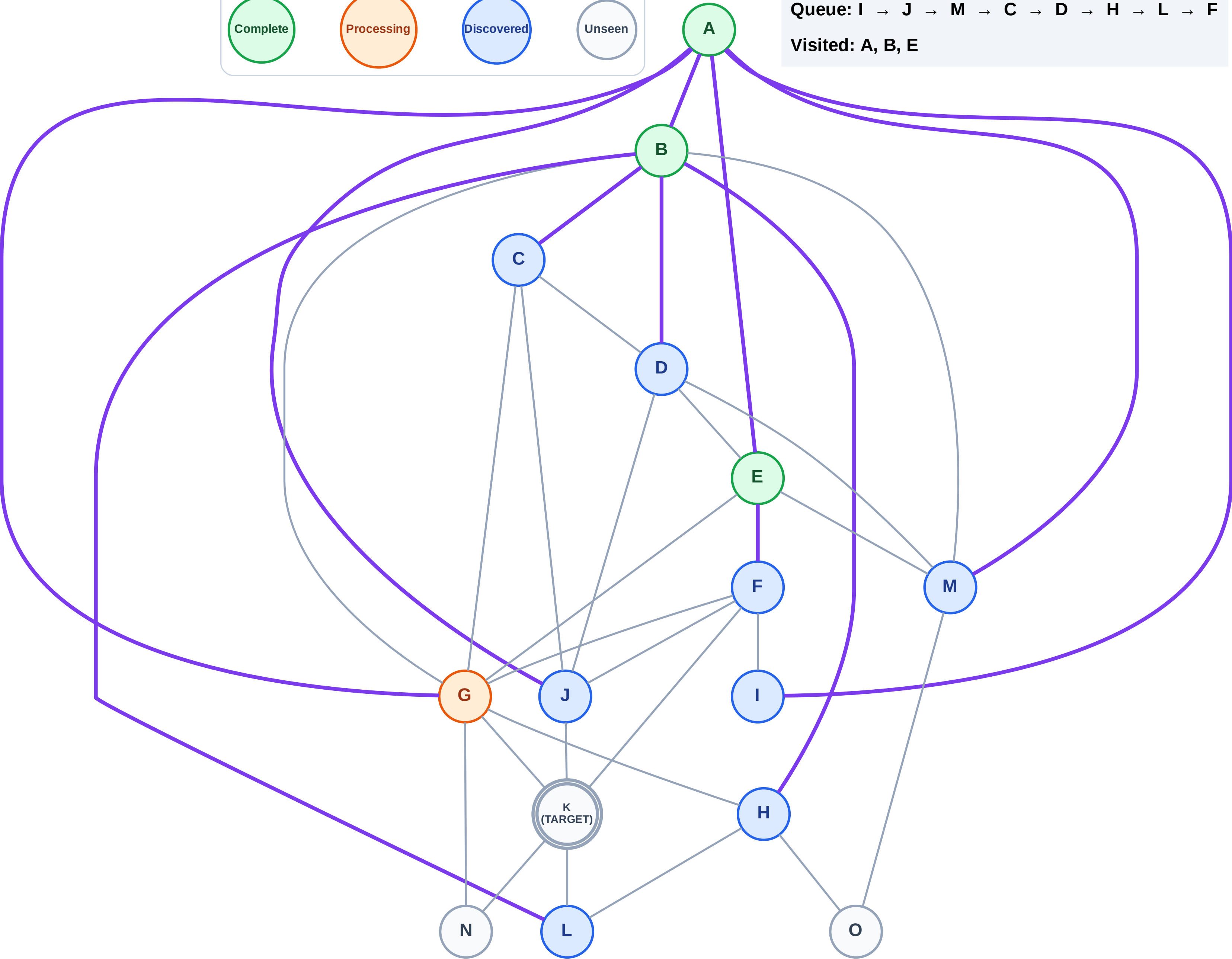
Processing

Discovered

Unseen

Queue: I → J → M → C → D → H → L → F

Visited: A, B, E



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → G → K

Legend

Complete

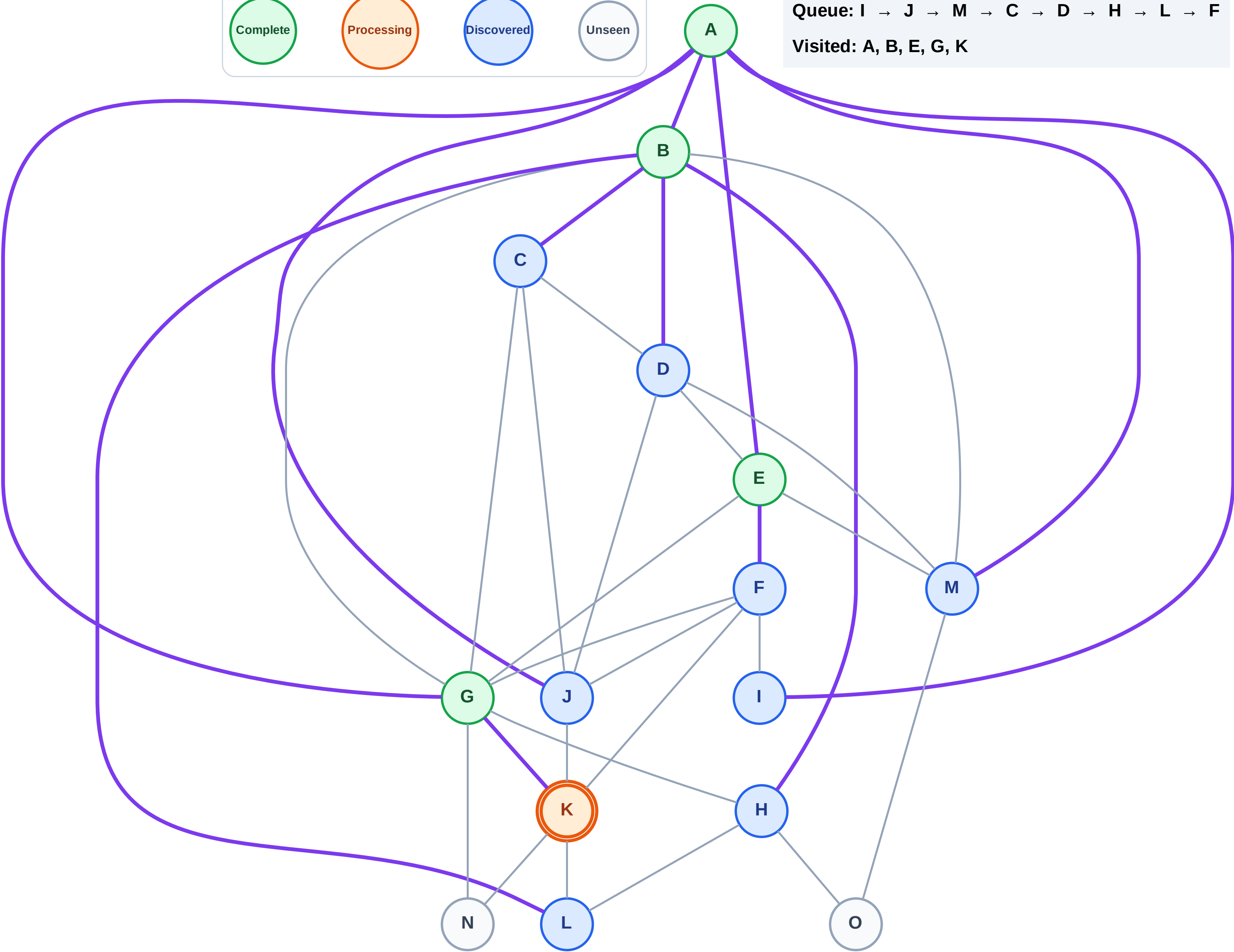
Processing

Discovered

Unseen

Queue: I → J → M → C → D → H → L → F

Visited: A, B, E, G, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

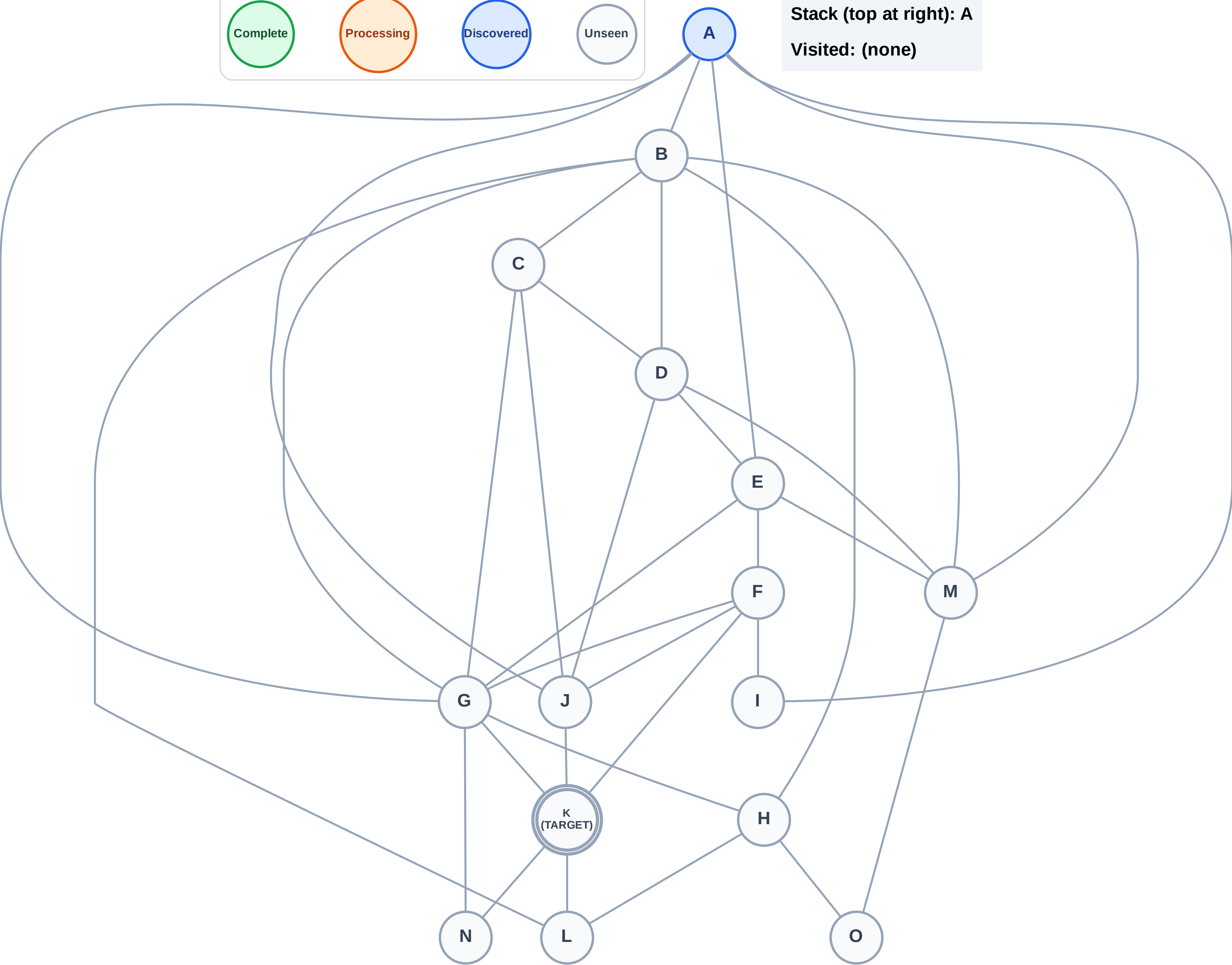
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

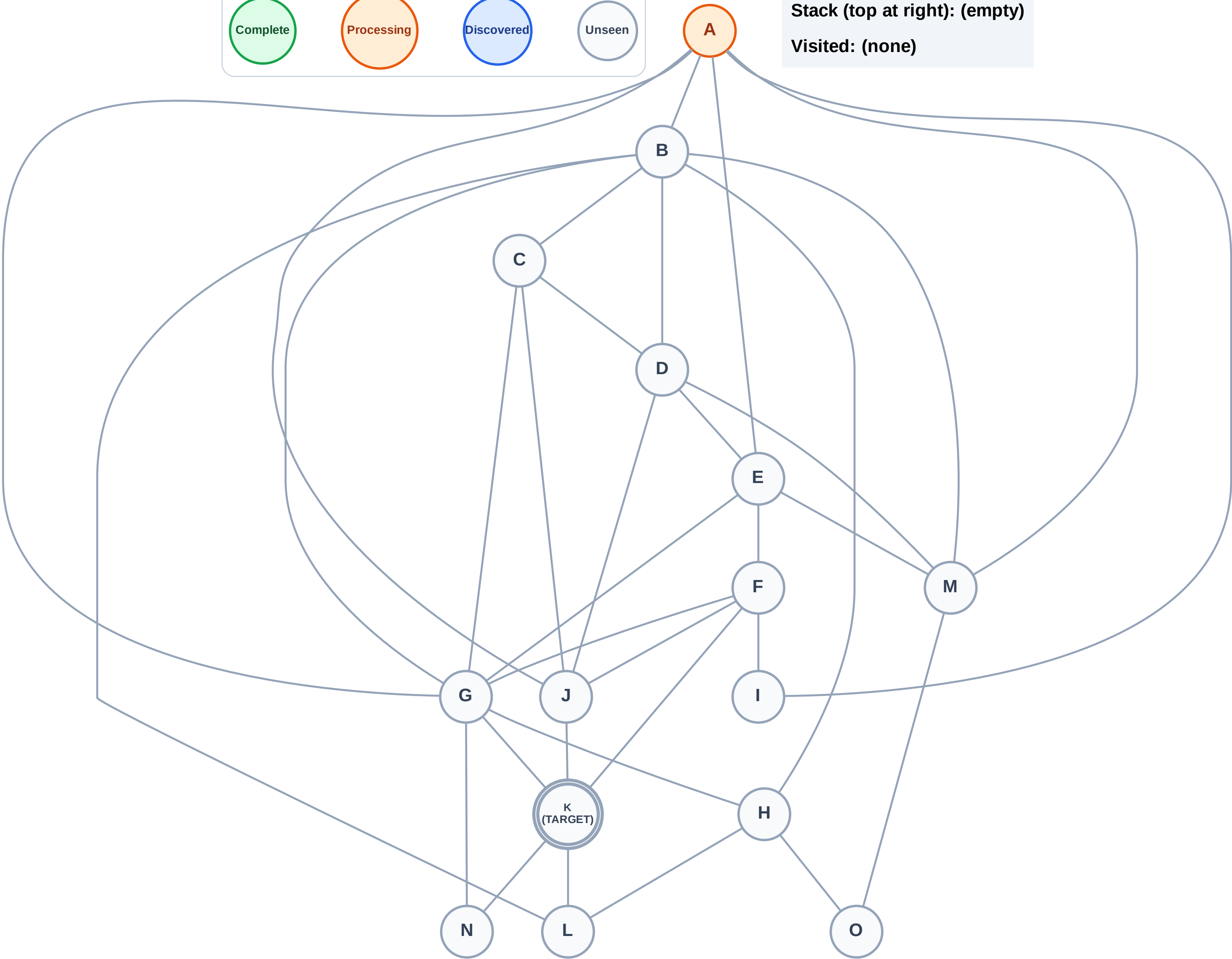
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

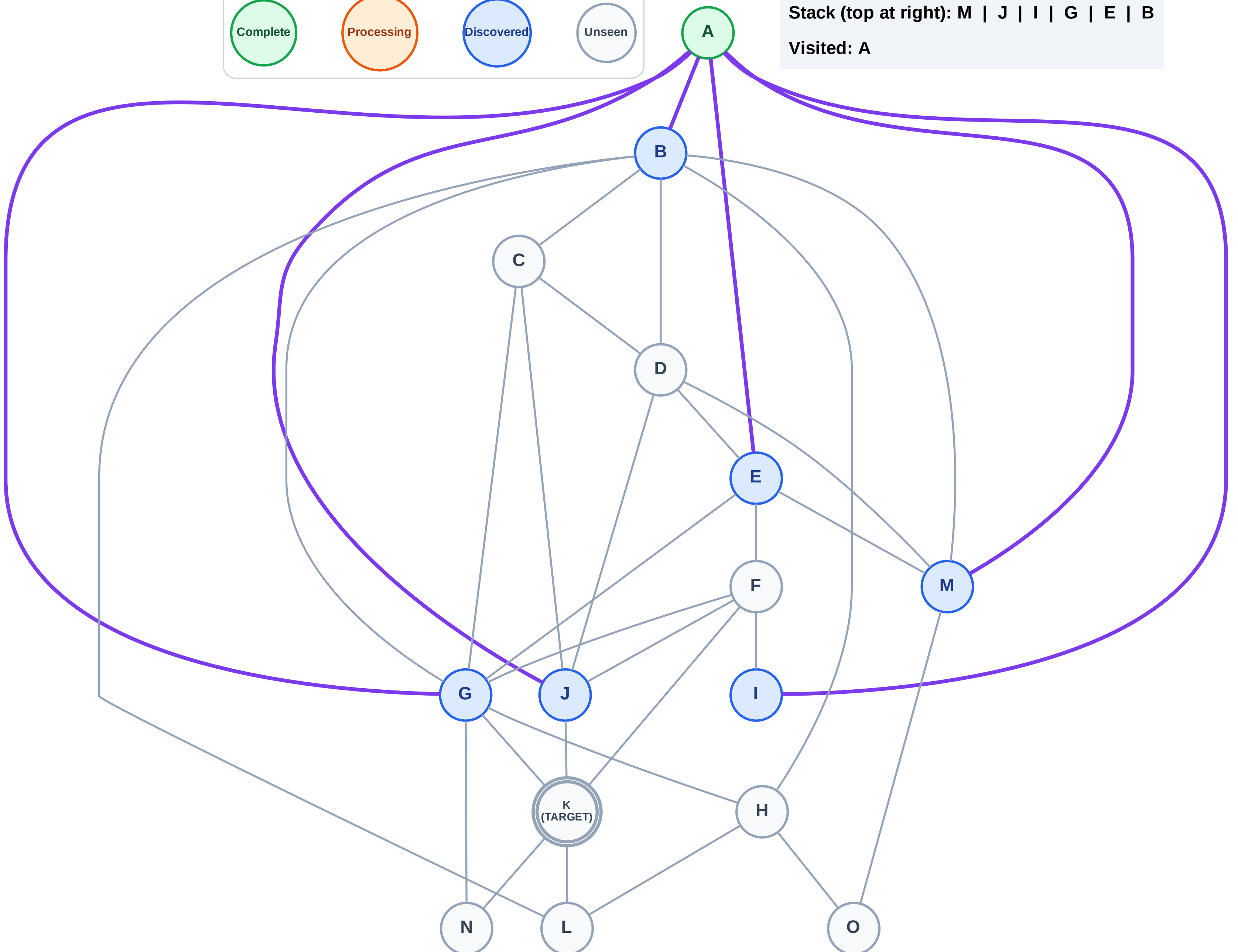
Step 3: Discover M, J, I, G, E, B from A

Legend



Stack (top at right): M | J | I | G | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

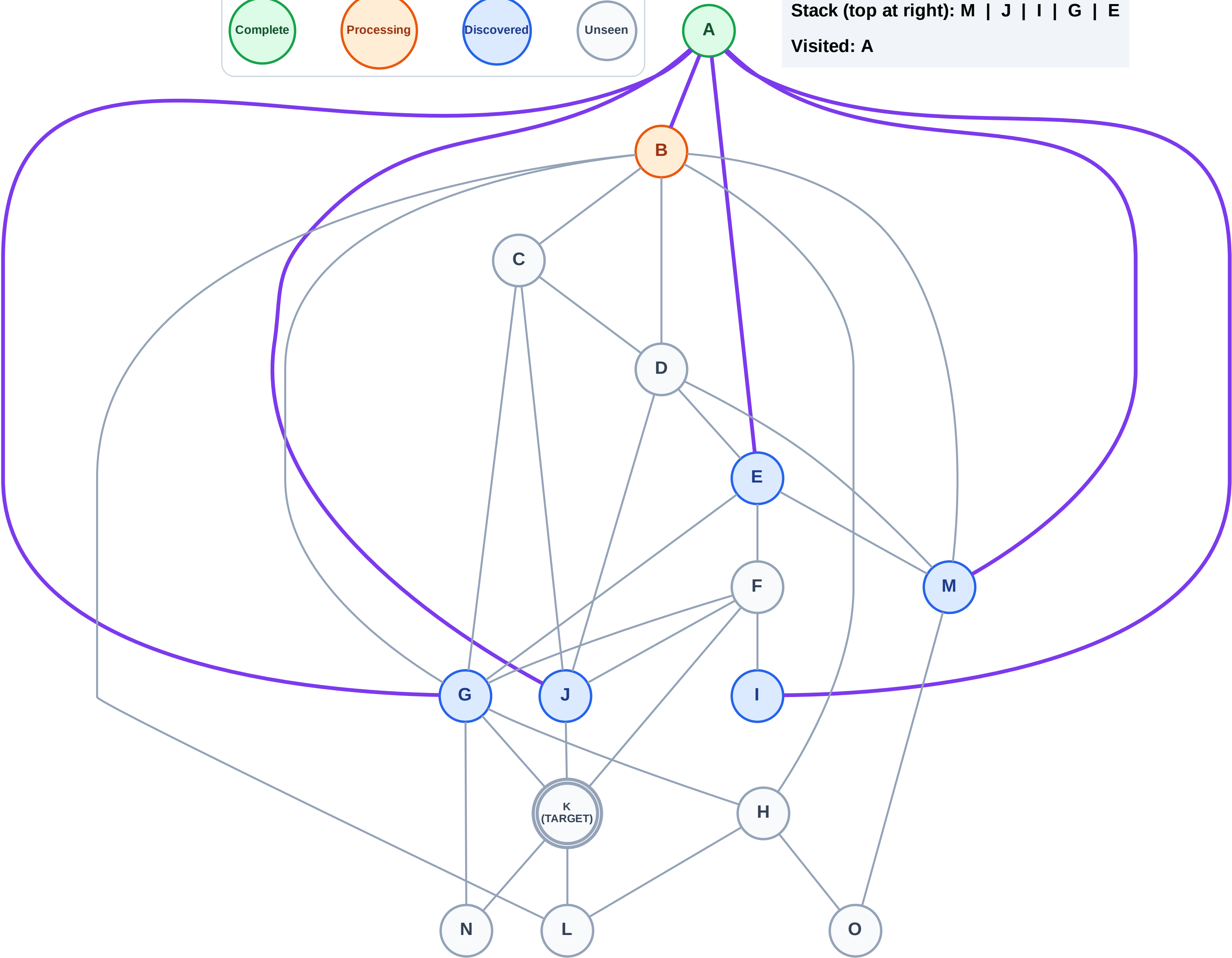
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E

Visited: A



DFS Traversal

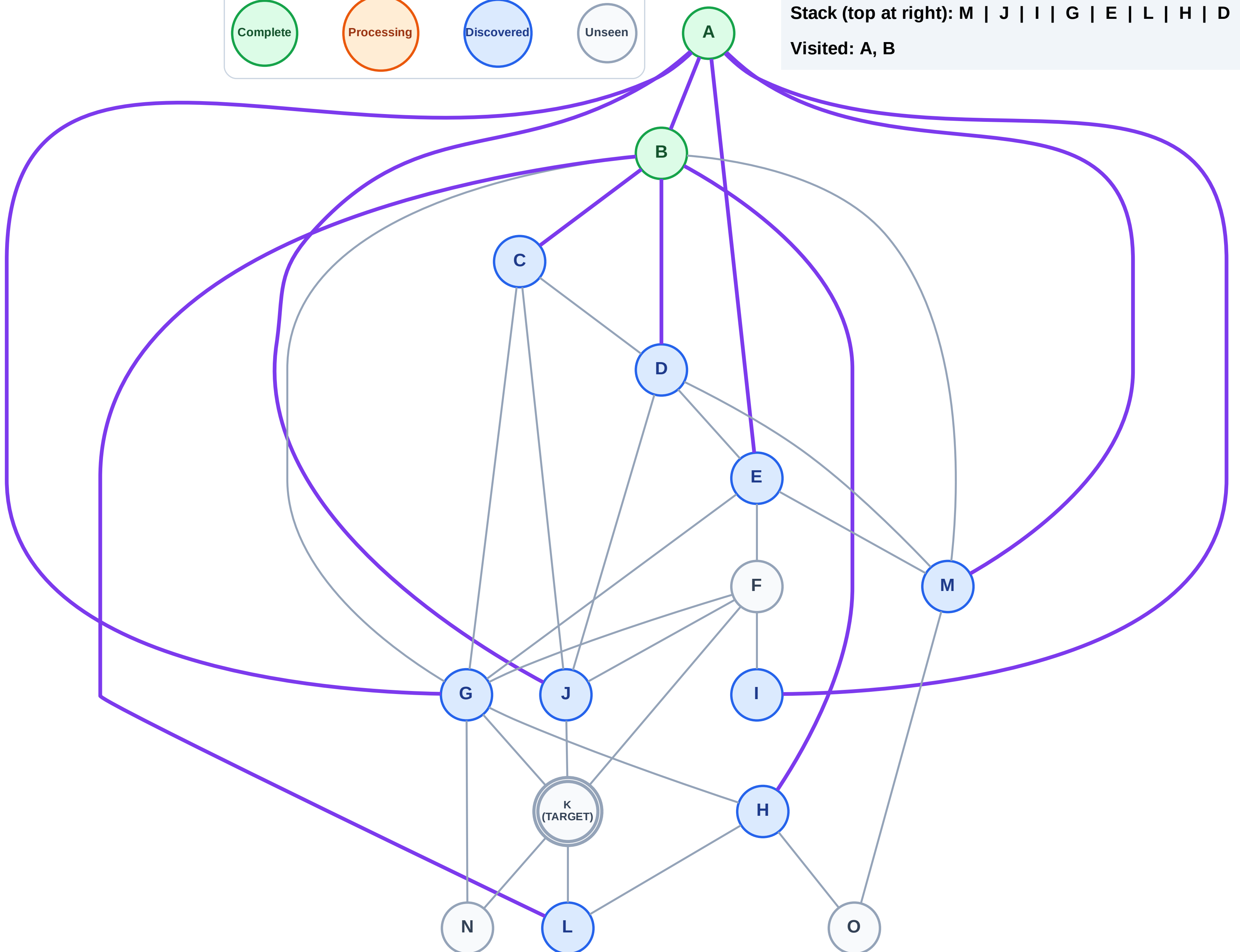
Step 5: Discover L, H, D, C from B

Legend



Stack (top at right): M | J | I | G | E | L | H | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

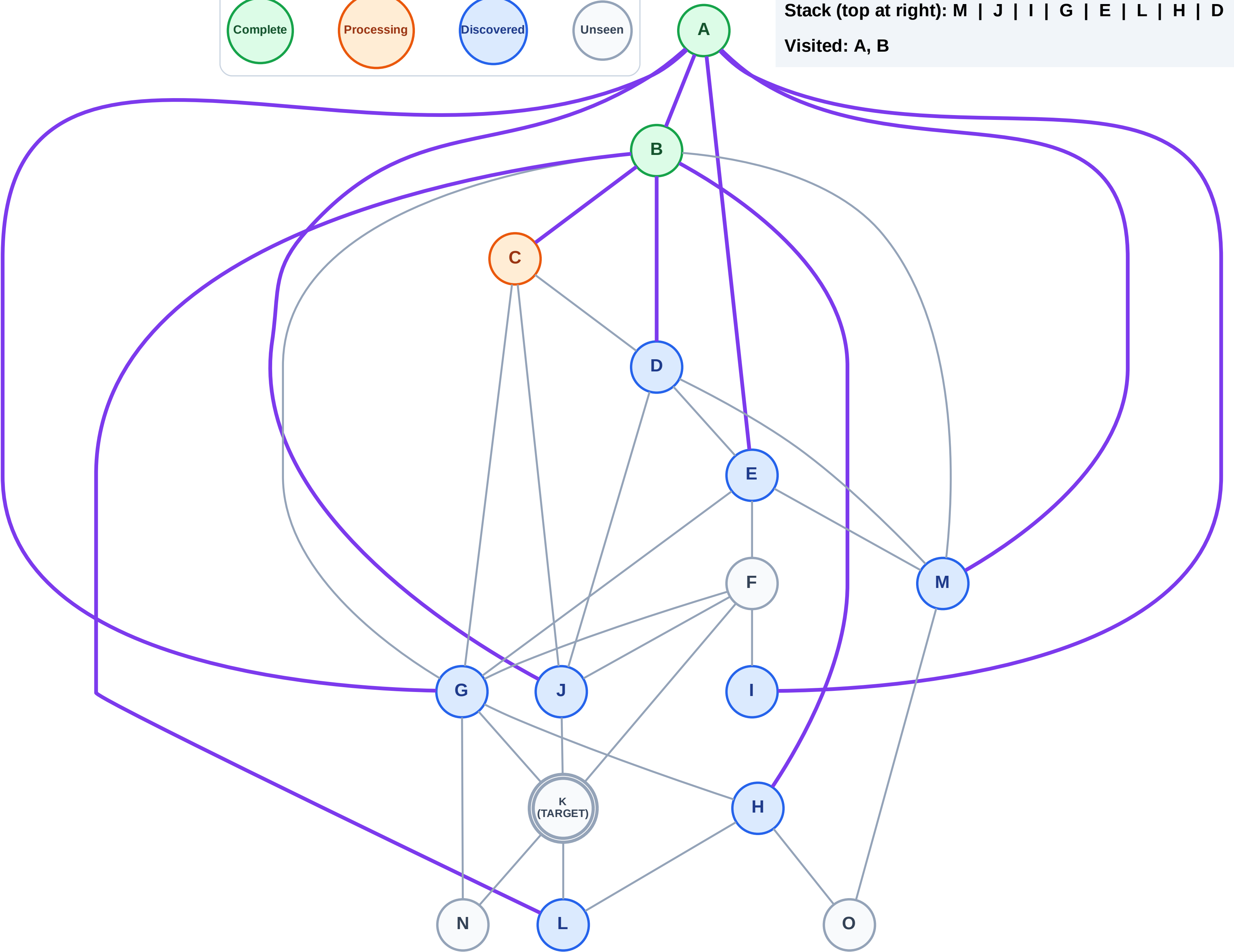
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E | L | H | D

Visited: A, B



DFS Traversal

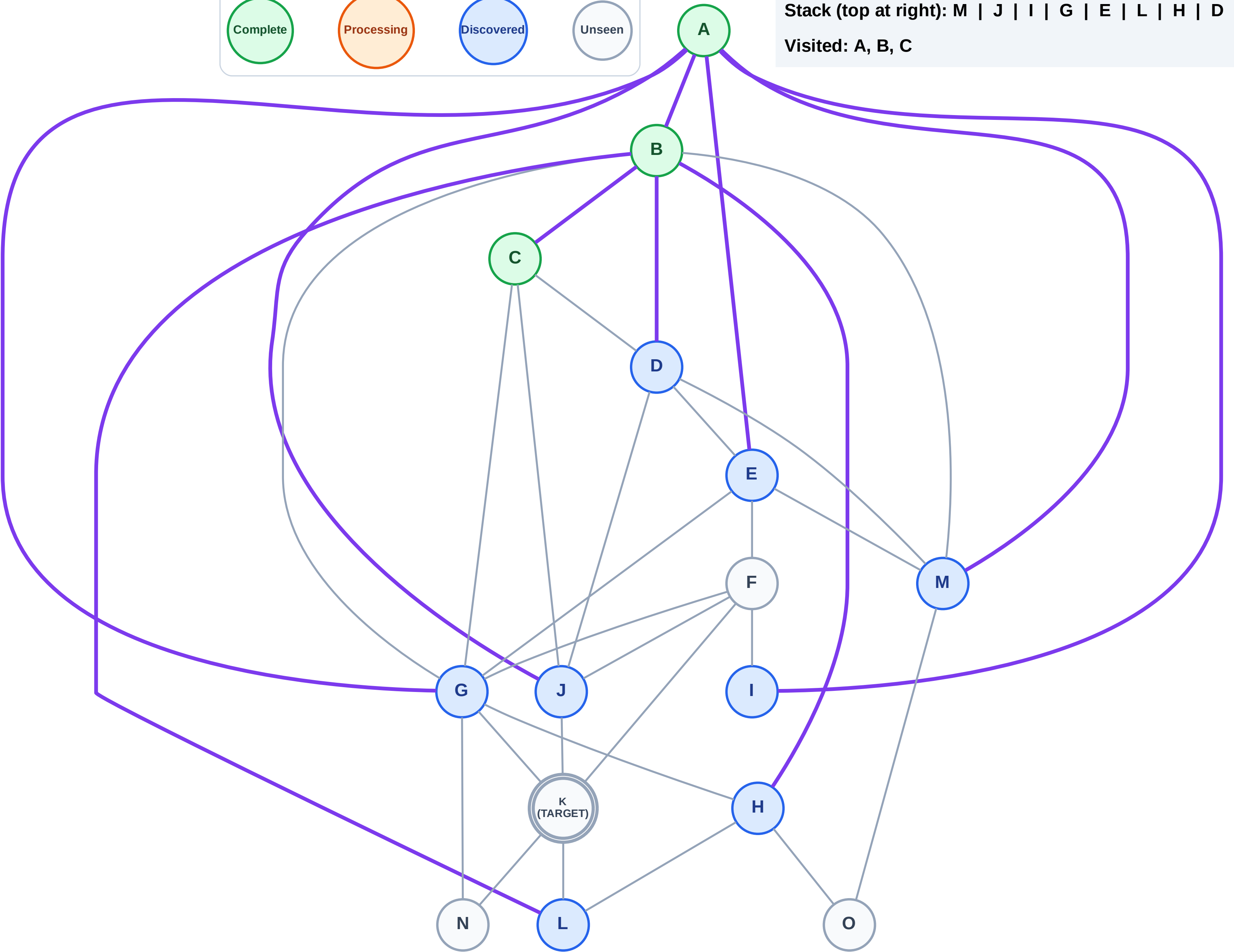
Step 7: No new neighbors from C

Legend



Stack (top at right): M | J | I | G | E | L | H | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

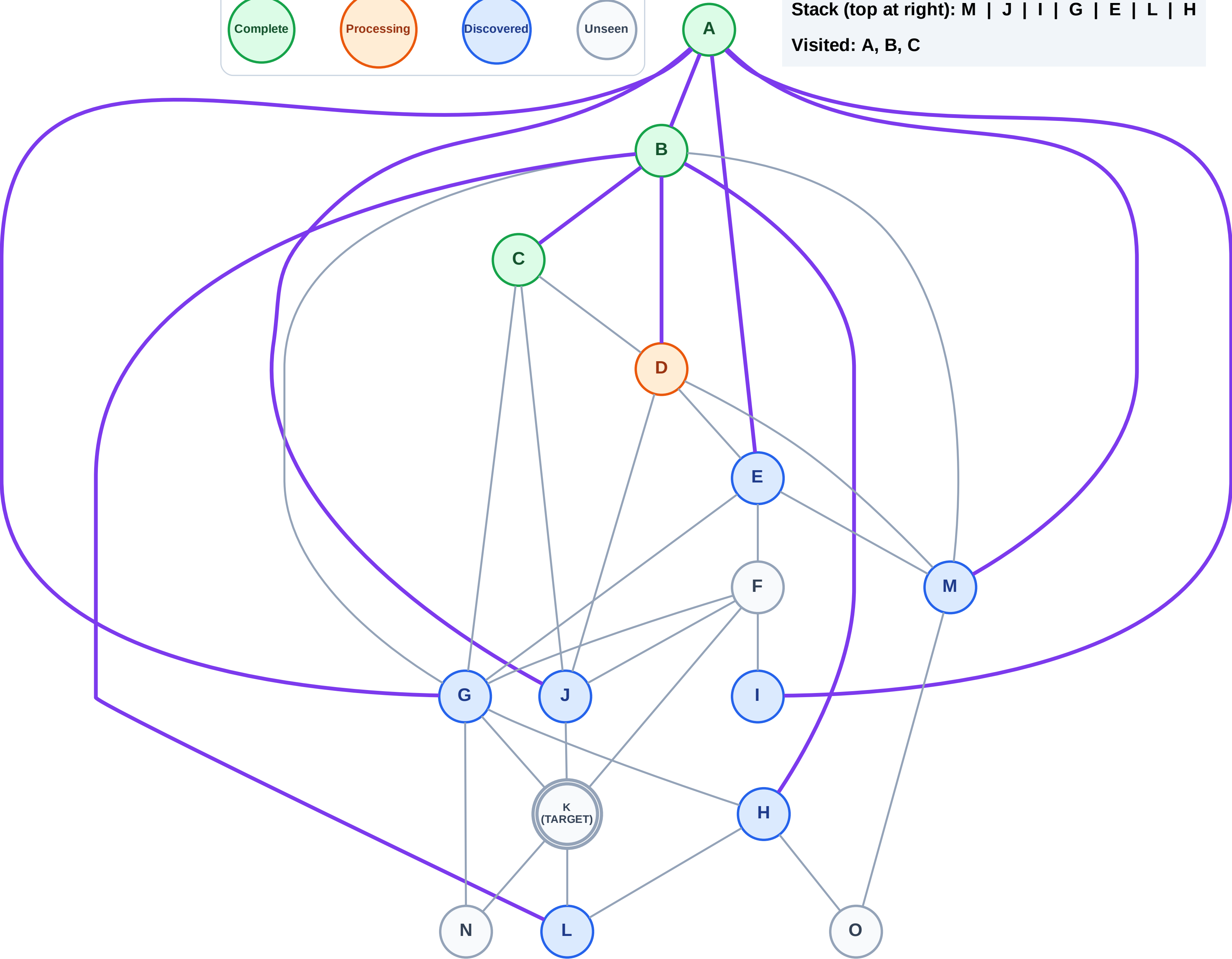
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E | L | H

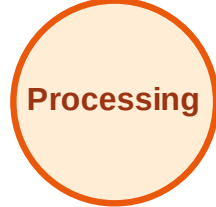
Visited: A, B, C



DFS Traversal

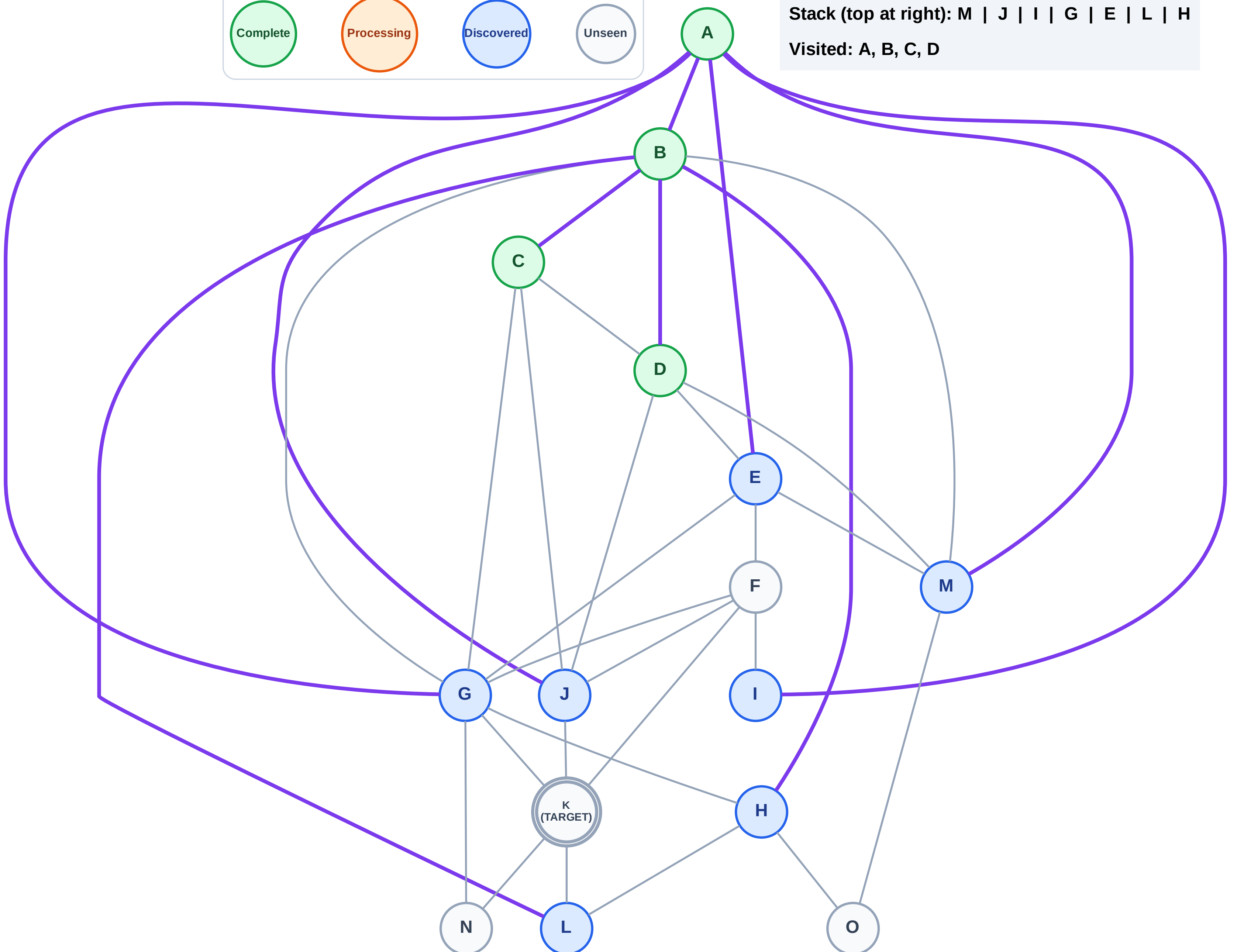
Step 9: No new neighbors from D

Legend



Stack (top at right): M | J | I | G | E | L | H

Visited: A, B, C, D



DFS Traversal

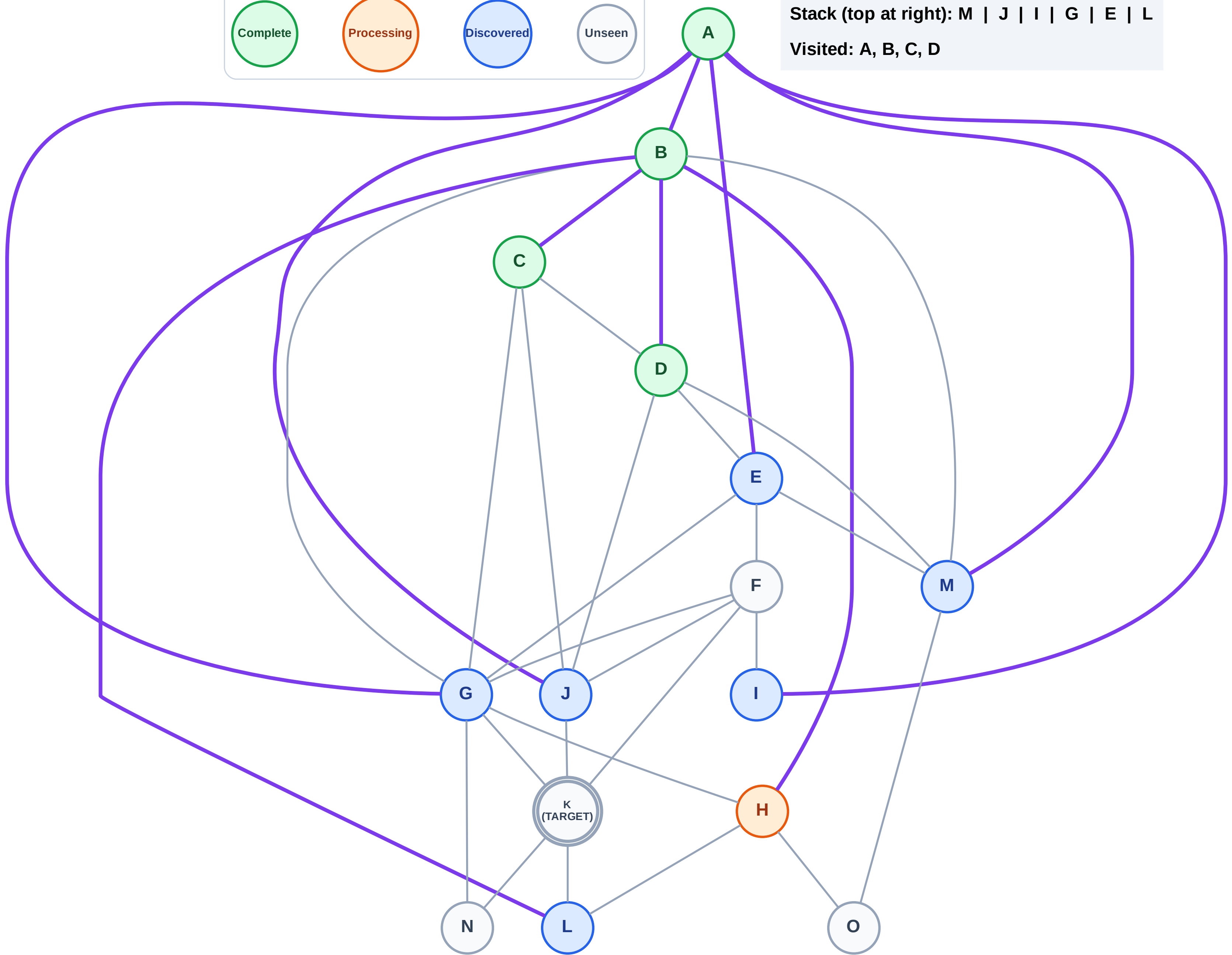
Step 10: Process node H

Legend



Stack (top at right): M | J | I | G | E | L

Visited: A, B, C, D



DFS Traversal

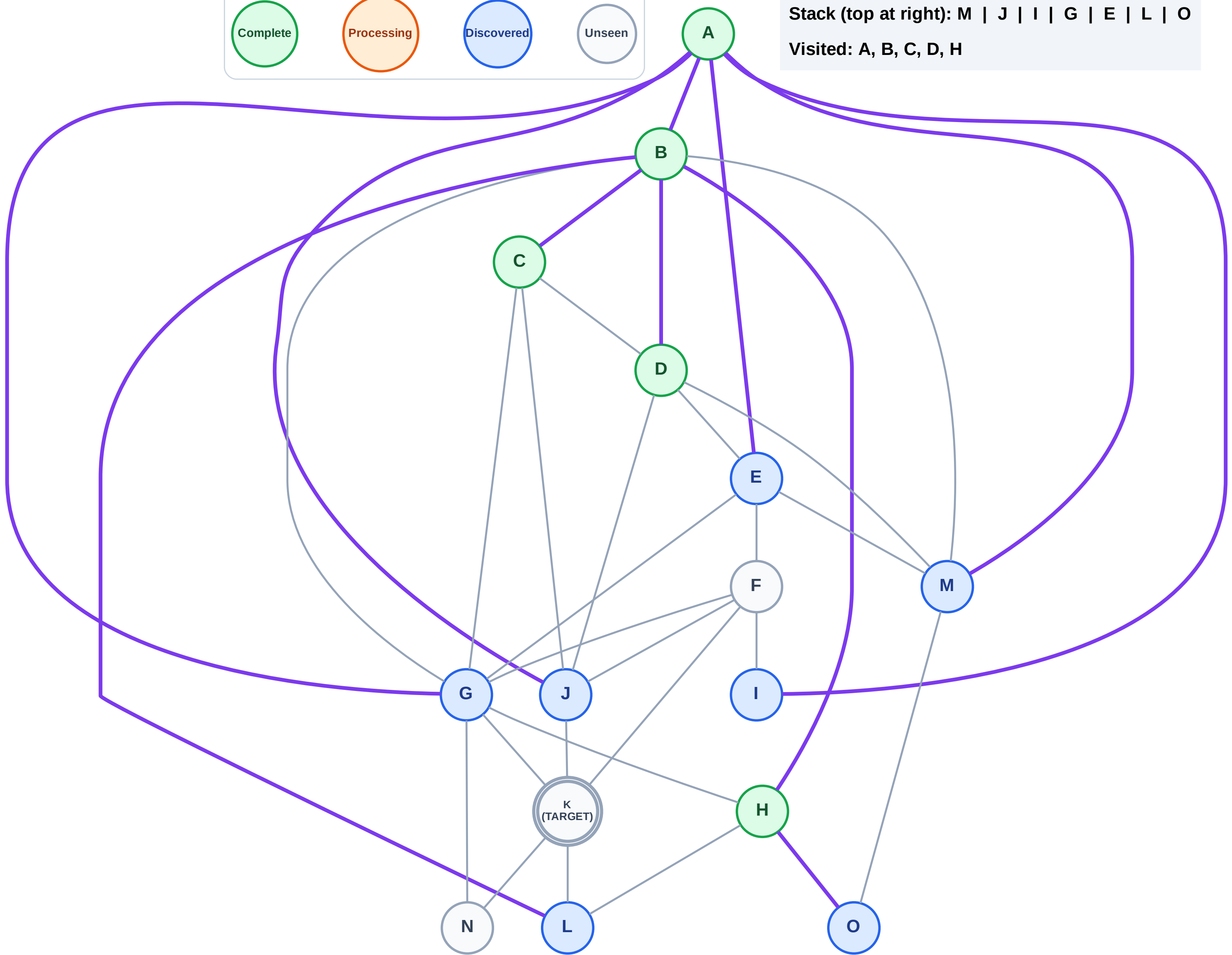
Step 11: Discover O from H

Legend



Stack (top at right): M | J | I | G | E | L | O

Visited: A, B, C, D, H



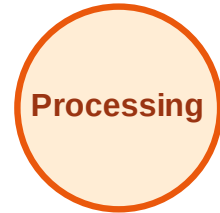
DFS Traversal

Step 12: Process node O

Legend



Complete



Processing



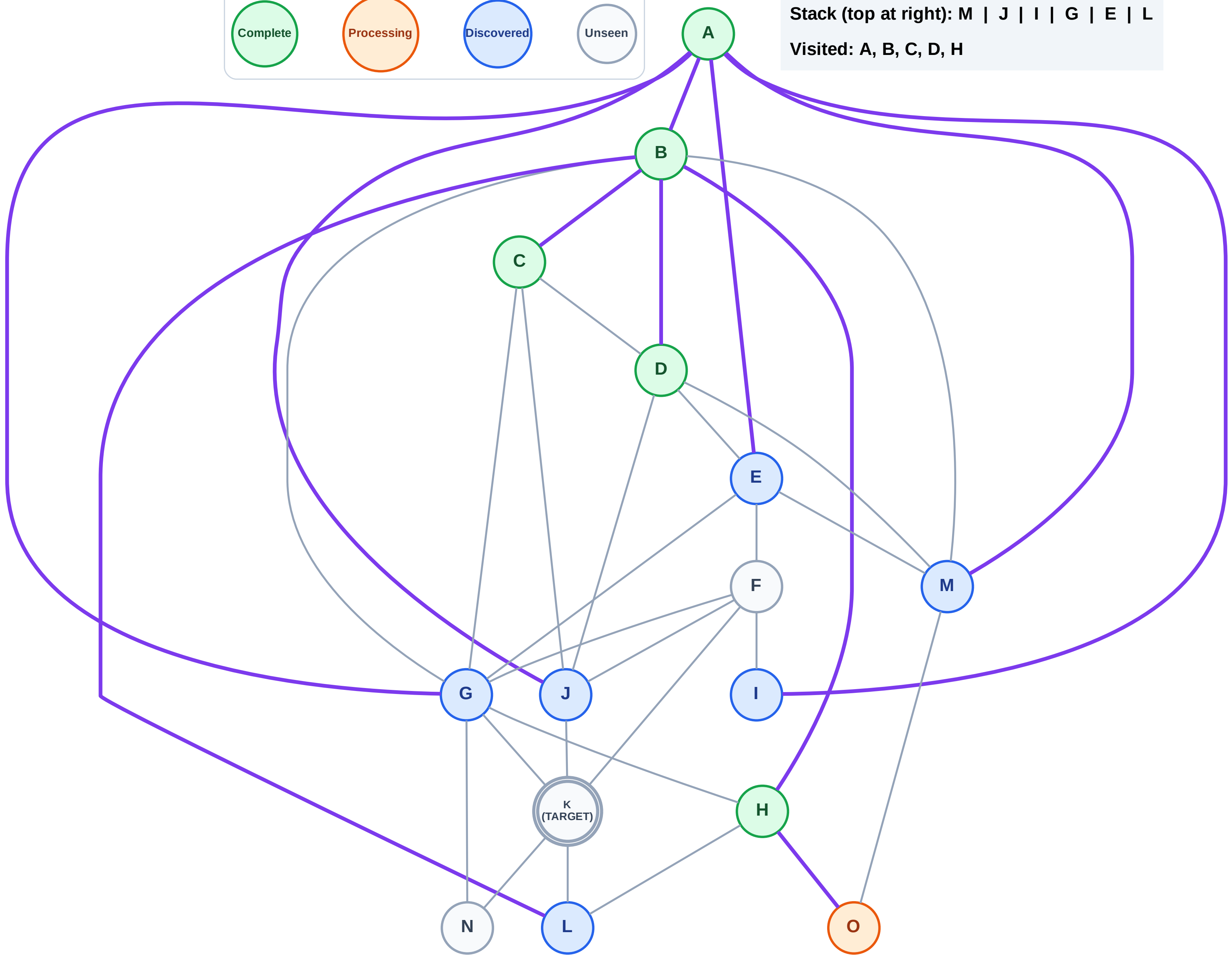
Discovered



Unseen

Stack (top at right): M | J | I | G | E | L

Visited: A, B, C, D, H



DFS Traversal

Step 13: No new neighbors from O

Legend

Complete

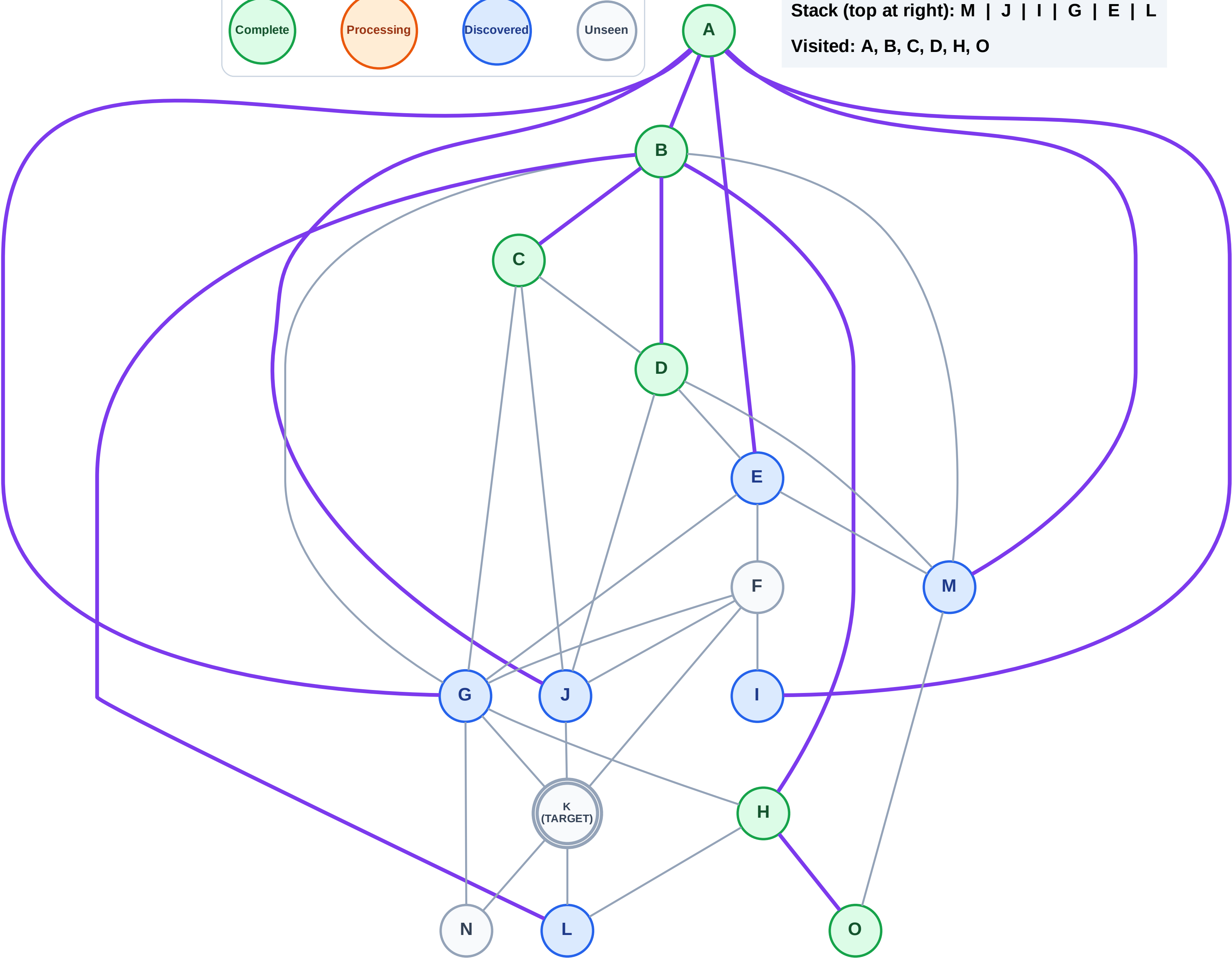
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E | L

Visited: A, B, C, D, H, O



DFS Traversal

Step 14: Process node L

Legend

Complete

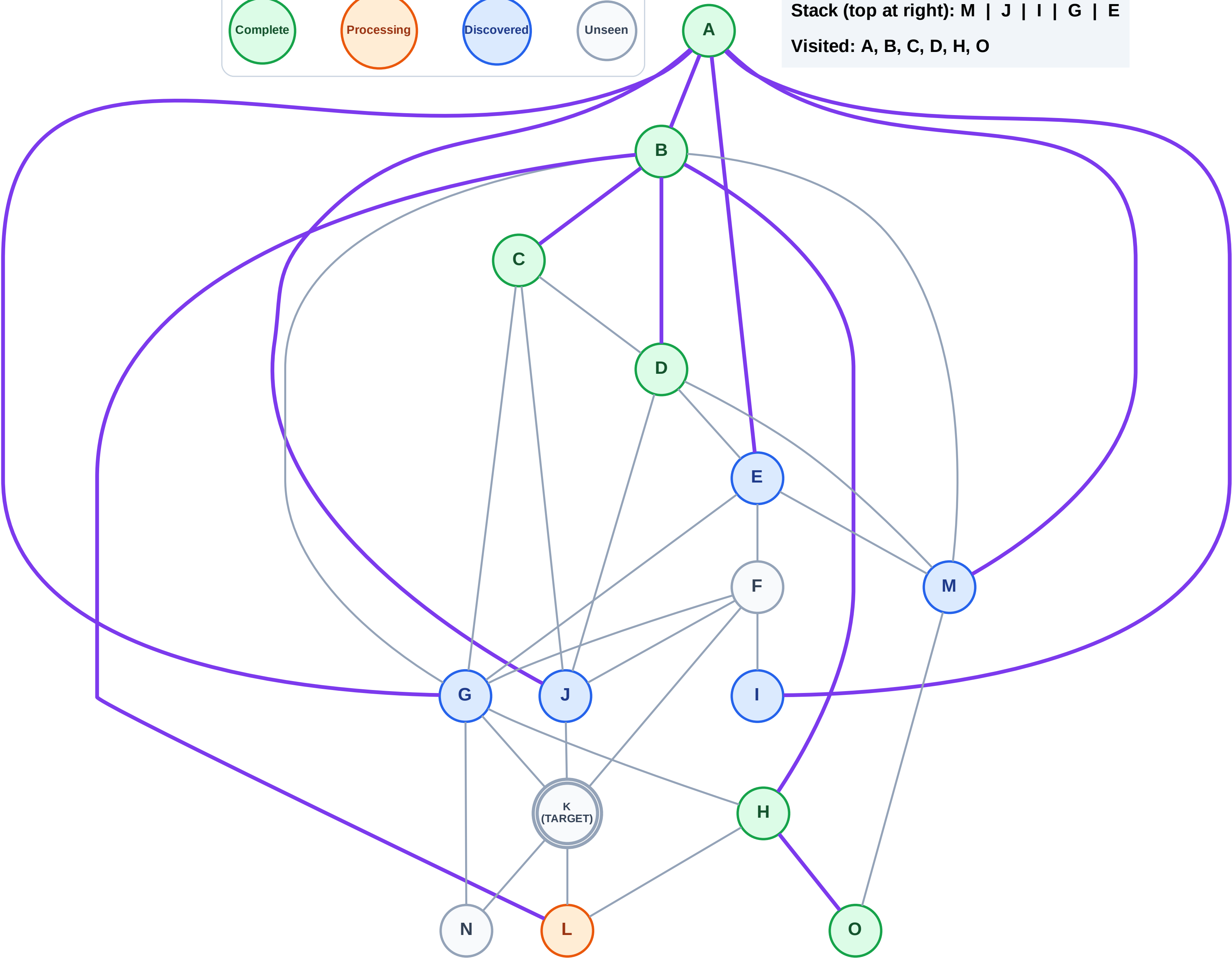
Processing

Discovered

Unseen

Stack (top at right): M | J | I | G | E

Visited: A, B, C, D, H, O



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → L → K

Legend



Stack (top at right): M | J | I | G | E

Visited: A, B, C, D, H, K, L, O

